

# Answer Key — Biology Grade 11

Sub-Strand 2.1: Reproduction in Plants • Lesson 8: Final Explanation: Why Cassava Cuttings Grow Without Seeds but a Lone Pawpaw Needs a Partner Tree

**Teacher reference only. Do not project or hand out.**

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| D  | D  | D  | B  | D  | C  | B  |

USE AFTER: PREDICT PHASE — Index card prediction: lone avocado in a Nairobi compound and a second avocado tree 10 m away

**Q1. A lone avocado tree in a Nairobi compound is covered in flowers but sets no fruit. A second avocado tree is planted 10 m away. Which outcome is most likely once both trees flower?**

- A. The second tree gives the first tree extra water, so its flowers will make fruit
- B. The first tree will start making fruit from its flowers by mitosis alone
- C. The two trees join their roots, so no pollen is needed to make fruit
- D. Pollen can be carried between the two trees, so cross-pollination and fertilisation can happen and fruit can form**

**Correct answer: D.** A lone tree may fail to fruit because of barriers to self-pollination. A second tree supplies compatible pollen, so cross-pollination, fertilisation and fruit development can follow.

USE AFTER: OBSERVE PHASE — Station A: vegetative propagation charts and photos of cassava, sugarcane and sweet potato cuttings

**Q2. A farmer in Kisumu plants a cassava stem cutting. It grows roots and shoots without any flower or pollen. Which statement explains this correctly?**

- A. The cutting needs pollen from another cassava plant before it can grow roots
- B. The cutting is a type of seed, so it grows like a seed
- C. Cells in the stem divide by meiosis, so the new plant is different from the parent
- D. Cells in the stem divide by mitosis to form roots and shoots, so the new plant is a clone of the parent**

**Correct answer: D.** Vegetative propagation uses mitosis in vegetative tissue. No gametes are involved, so the offspring is genetically identical to the parent.

USE AFTER: OBSERVE PHASE — Station B: labelled flower diagrams and classification tables, including features that hinder self-pollination

**Q3. Which of these is a feature that hinders self-pollination in flowering plants?**

- A. Anthers and stigma of the same flower ripen at the same time and touch each other
- B. The flower produces a lot of nectar
- C. The flower has a large ovary with many ovules
- D. The stigma of a hibiscus flower matures at a different time from its anthers**

**Correct answer: D.** Different times of maturity of anthers and stigma (dichogamy) make it hard for a flower to pollinate itself. Nectar and a large ovary do not prevent self-pollination.

USE AFTER: OBSERVE PHASE — Station C: double fertilisation animation screenshots and group summary of zygote and endosperm formation

**Q4. During double fertilisation in a flowering plant such as maize, what happens to the two sperm nuclei from the pollen tube?**

- A. One sperm nucleus forms the seed coat, and the other forms the fruit wall
- B. One sperm nucleus fuses with the egg cell to form the zygote, and the other fuses with the polar nuclei to form the endosperm**
- C. Both sperm nuclei fuse with the polar nuclei to form the endosperm
- D. Both sperm nuclei fuse with the egg cell to form two zygotes

**Correct answer: B.** One sperm nucleus fertilises the egg to make the diploid zygote. The second joins the polar nuclei to make the endosperm, which feeds the embryo.

USE AFTER: OBSERVE PHASE — Station D: fruit sections of tomato, passion fruit and bean pod with placentation and fruit type classification

**Q5. A learner cuts open a tomato fruit and finds small seeds inside. Which statement about how the fruit and seeds formed is correct?**

- A. The petals became the fruit and the anthers became the seeds
- B. The ovule became the fruit and the ovary became the seeds
- C. The whole flower became the fruit and the seeds came from the stigma
- D. The ovary became the fruit and the ovules became the seeds**

**Correct answer: D.** After fertilisation, ovules develop into seeds and the ovary wall develops into the fruit.

USE AFTER: DQB PHASE — Sorting DQB sticky notes into 'Answered', 'Partly answered' and 'New question', including unresolved confusion between pollination and fertilisation

**Q6. A student looks at a pawpaw flower and sees pollen grains on the stigma. Which conclusion is correct?**

- A. Neither pollination nor fertilisation can happen without a second tree
- B. Pollination and fertilisation mean the same thing, so the fruit will form
- C. Pollination has happened, but fertilisation has not necessarily happened yet**
- D. Fertilisation has definitely happened, because pollen is on the stigma

**Correct answer: C.** Pollination is the transfer of pollen from anther to stigma. Fertilisation happens later, when the male gamete fuses with the egg, and only if the pollen is compatible and a pollen tube grows.

USE AFTER: MODEL BUILDING PHASE — Final model and unit assessment tasks: advantages and disadvantages of asexual and sexual reproduction

**Q7. A sugarcane farmer in Mumias plants setts (stem pieces) instead of seeds. What is an advantage and a disadvantage of this method?**

- A. Advantage: it needs double fertilisation. Disadvantage: it makes fewer plants
- B. Advantage: the crop grows fast and is uniform. Disadvantage: the plants are clones, so a disease can spread to all of them**
- C. Advantage: it needs pollen from another farm. Disadvantage: it makes the plants different from each other
- D. Advantage: it creates a lot of genetic variation. Disadvantage: it needs pollinators

**Correct answer: B.** Setts produce clones by mitosis. This gives fast, uniform crops, but the lack of variation means disease can affect all plants.